N. Martin Mbithi

Software Engineer | Cybersecurity and Digital Forensics | Cloud Architect

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PROFILE

Software engineer with eight years of experience in backend development, systems integration and database-driven applications, and a background in cybersecurity and digital forensics (MSc, Open University of Kenya). At the Government of Makueni County I am part of the team behind iGrid017, the county's integrated digital platform: I work across its core ERP, build connected systems and integrations, and manage the county's cloud services. I apply security practice to what I build, from role-based access and server-side validation to system audits, and previously ran security audits and controls as an ICT Officer with the ICT Authority.

CORE COMPETENCIES

Core: PHP, Laravel, MySQL, REST API design, Linux, Nginx, Git

Cybersecurity and forensics: Secure software development, authentication and authorisation, role-based access control, TLS and CSRF protection, system audits, digital forensics

Cloud and infrastructure: Cloud services management, Linux and Unix administration, Nginx, Docker, Kubernetes, CI/CD, AWS, Google Cloud

Integration: Bank and payment gateway APIs, SMS and email gateways, Swagger, Postman

Delivery: Code review, automated testing, Agile and Scrum, JIRA, GitHub, GitLab

Also used: Python (Django, Flask), Java (Spring Boot), JavaScript (Node.js, React, Vue), PostgreSQL, MongoDB, Redis

PROFESSIONAL EXPERIENCE

Software Engineer, Applications Development

Mar 2023 to present

Government of Makueni County

Part of the in-house team behind iGrid017, the county's integrated digital platform, with the county ERP (launched August 2026) at its core and departmental systems and public e-services connected to it. I work across the platform and manage the county's cloud services.

- **ERP platform and modules:** work across the iGrid017 platform and its modules in public finance, public service management, planning, health, procurement, participatory development and county programmes; in public finance, build and maintain the backend services and data models for revenue billing and receipting, programme-based budget formulation and absorption tracking, and expenditure requisitions through vouchers to payment.
- **Cloud services:** manage the county's cloud services, on Linux and Nginx.
- **Integrations:** design and maintain the REST integrations between county systems and banks, payment gateways and SMS and email services.
- **Databases and county systems:** optimise schemas and queries behind cross-module reports; delivered linked county systems including the procurement portal, PMTS, IVAS, Kilimo Makueni and the ECDE, VTC and CIC systems.
- **Engineering practice:** introduced a Git-based workflow, code review and automated testing in the team; mentor junior engineers and work with finance, procurement and operations staff to turn policy requirements into working software.

ICT Officer

Jan 2022 to Feb 2023

ICT Authority, Machakos County Headquarters

- Ran regular system audits and implemented cybersecurity controls against data breaches and unauthorised access, alongside network and user support.

Mobile Application Developer Jan 2021 to Nov 2021
World Christian Fellowship (NGO)

- Built native iOS and Android apps in Swift and Java with geolocation and social sharing integrations; memory profiling cut load times by 20%.

Web Application Developer Jan 2020 to Jul 2021
Ezana Group

- Built server-side components and REST APIs in Node.js and Express, and responsive front ends in React and Vue.

Web Developer and ERP Support Officer Jan 2020 to Jan 2021
Ukamba Bible College

- Supported and extended the college ERP and website; reduced page load times, cutting bounce rate by 15%.

Software Engineer Jan 2018 to Jan 2020
DevLan Solutions Ltd

- Introduced code review, Git version control and unit testing across the team, and built REST APIs for data exchange between components.

Graphic Designer Jan 2016 to Jan 2018
Lauh Print Company, Machakos

EDUCATION

MSc, Cyber Security and Digital Forensics, Open University of Kenya

BSc, Computer Science, St Paul's University 2017 to 2020

CERTIFICATIONS

- Software Engineering Certification, Open University of Kenya, 2025
- Agile Project Management and JIRA, HP Foundation, Atlassian and Great Learning, 2024
- DevOps Certification, Edureka, 2022
- Cyber Security Essentials, Cisco, 2019
- Mobile Web Specialist, Andela and Google Africa Developer Scholarship, 2018
- NDG Linux Unhatched and Linux Essentials, Cisco, 2017